Tutorial 9: Linear regression

1. Consider the set of paired data $(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)$ to which we fit the linear regression model:
$$Y_i \sim N(\alpha + \beta x_i, \sigma^2),$$
where the Y_i are independent. Show that the residual sum of squares $RSS = \sum(y_i - \hat{\alpha} - \hat{\beta}x_i)^2$ can be calculated from the summary statistics as $RSS = s_{yy} - \frac{s_{xy}^2}{s_{xx}}$.

2. The relationship between the cost of flood damage and the depth of flooding in two storey dwellings is investigated using the following sample of flood insurance claims:

x :	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6
y :	10	14	26	28	29	41	43	44	45	46	47	48	49

where x is the (standardised) depth of flooding, in feet, above first floor level and y is the flood damage as a percentage of structure value.

   (a) Draw a scatter plot of the data and comment on the form of the relationship between y and x.

   (b) Fit the regression model $\mathbb{E}(Y|x) = \alpha + \beta x$ to these data.

   (c) By making suitable assumptions, which should be stated, calculate a 95% confidence interval for β.

   (d) Construct a table of residuals from the fit in (a) and demonstrate graphically any reservations you may hold about the suitability of this model.

3. Consider the data in Question 2 of Tutorial 8.

   (a) Draw a scatter plot of the data and comment briefly.

   (b) Derive the equation of the line of best fit (y on x).

   (c) Find a 95% confidence interval for the slope of the regression line, β. Based on this interval, test the hypothesis that there is no linear relationship between the amount of payment and the amount of claim. What is the significance level for the test that you have performed?

   (d) Calculate (i) a 95% confidence interval for the mean payment on claims of £780, and (ii) a 95% prediction interval for the actual individual payment on a claim of £780.

   (e) Calculate R^2 and from that compute the correlation coefficient r. Comment on their values.

4. In an experiment to investigate how the growth of a species of bacteria is influenced by glucose concentration ($x, mmol\, l^{-1}$), 8 cultures are grown in different concentrations of glucose and logarithms of the numbers of bacteria (y) in each culture are estimated. The results are given below.

Gluc. conc.	log bacteria count
2.5	10.6
3.0	11.6
3.5	15.1
4.0	15.2
4.5	17.8
5.0	20.4
5.5	22.1
6.0	26.0

For these data $\sum x = 34$, $\sum x^2 = 155$, $\sum y = 138.8$, $\sum y^2 = 2603.38$, $\sum xy = 634.6$.

(a) Fit the regression model

$$Y_i = \alpha + \beta x_i + \varepsilon_i,$$

to these data and calculate a 95% confidence interval for the slope parameter β.

(b) Plot the data. How appropriate do you think the linear model is?

(c) What is the predicted value of bacteria count when the glucose concentration is 3.7 ($mmol\, l^{-1}$)?

(d) How much do you expect the bacteria count to change for every additional one-unit increase of the glucose concentration?

Exam Style Question

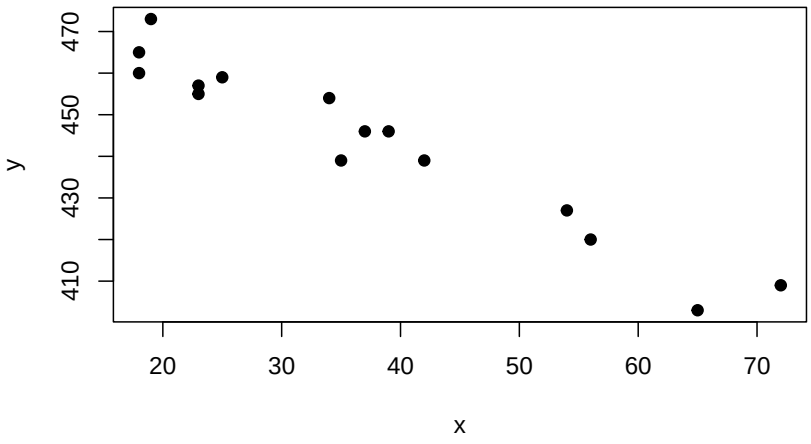
A recent study suggests that the sleeping time, in minutes, (y) required by adults according to their age, in years (x) is given by the equation,

$$\text{Sleeping Time} = 480 – \text{Age}.$$

Suppose this is to be empirically checked and 15 people of varying ages are asked to report how long they slept on a particular standard night. The following data are collected:

x	18	23	25	35	65	54	34	56	72	19	23	42	18	39	37
y	460	457	459	439	403	427	454	420	409	473	455	439	465	446	446

(a) By referring to the scatter diagram below, comment on the relationship between age and sleeping time.



[2 marks]

(b) Given that

$$\sum x_i = 560, \sum y_i = 6652, \sum x_i^2 = 25188, \sum y_i^2 = 2955918, \sum x_i y_i = 243444$$

calculate the fitted line for the regression of y on x.

[3 marks]

(c) Assuming that we are dealing with a full normal model of regression, find a 95% confidence interval for an individual value for the number of hours sleep taken by a 60 year old person.

[3 marks]

(d) Does your answer to (c) seem to support the claim of the study that Sleeping Time = 480 − Age?

[2 marks]

[Total 12 marks]